

Ella Yee

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EDUCATION

Columbia University in the City of New York

New York, NY

B.S. in Biomedical Engineering, Minor in Computer Science

September 2024 - May 2028

- Awards: Egleston Scholar (Top 1% of Engineering Applicants), SURF Fellow, Dean's List (all semesters)
- Coursework: Artificial Intelligence, Computing for Engineers (Python), Data Structures (Java), Advanced Programming (bash, C), Discrete Math, Multivariable Calculus, Differential Equations, Linear Algebra, Biology
- GPA: 4.2/4.0

The Harker School

San Jose, CA

High School Diploma

August 2020 - May 2024

- Coursework: Neural Networks, Data Structures, AP Computer Science A, AP Statistics, Compilers & Interpreters, AP Calculus BC, Multivariable Calculus, Differential Equations, Linear Algebra, Discrete Math, Honors Biology
- GPA: 4.0/4.0, SAT: 1600/1600

SKILLS

- **Languages & Tools:** Python (PyTorch, NumPy, pandas, SciPy), R (DESeq2, edgeR), Java, Unix, Git, Excel
- **Computational Biology:** RNA-seq analysis (bulk, single-cell, spatial), machine learning, statistical modeling
- **Molecular Biology:** PCR, DNA purification, electrophoresis, immunohistochemistry, histology, cell culture

EXPERIENCE

Columbia Engineering

New York, NY

Research Intern, Azizi Lab (Computational Cancer Biology Laboratory)

September 2025 - Present

- Apply deep generative modeling (Python) to single-cell and spatial RNA-seq datasets of graft-versus-host disease
- Develop workflows for high-dimensional data visualization and inference, lead cross-lab communications

Columbia Summer Undergraduate Research Fellowship (SURF)

New York, NY

Research Fellow, Abate-Shen Lab

May 2025 - August 2025

- Performed computational and immunohistological analysis of prostate cancer bone metastasis in mouse models
- Evaluated 20+ organ-on-a-chip platforms as ex vivo cancer models, identified biological/engineering constraints

Columbia University Irving Medical Center

New York, NY

Research Intern, Abate-Shen Lab

September 2024 - May 2025

- Conducted differential expression analyses (R) of RNA-seq datasets from genetically-engineered mouse models

Stanford Cancer Institute

Stanford, CA

Research Intern, Sunwoo Lab

June - August 2024, June - August 2023

- Performed molecular cloning, DNA purification, and PCR to evaluate CAR-NK cancer immunotherapy
- Led independent project on CRISPR-Cas9-mediated gene knockout, performed quantitative data analysis

NewMind Discovery & Harker Labs

San Jose, CA

Independent Researcher

August 2020 - May 2024

- Led 4 independent projects in computational biology and machine learning, using large-scale biological datasets
- Designed scalable deep learning models, statistical modeling, and image analysis pipelines in Python and R

UC Davis Young Scholars Program (YSP)

Davis, CA

Research Scholar, Korf Lab

June - July 2022

- Participated in 6-week residential research program (<12% acceptance rate) at UC Davis Genome Center
- Developed algorithms to identify ultraconserved genomic elements (UCEs) across 4 plant species through integration of Python, Unix, JBrowse, and NIH BLAST, named "Summer Slam" awardee for best presentation

PUBLICATIONS & AWARDS

- Columbia University Egleston Scholar, Summer Undergraduate Research Fellow, Dean's List (all semesters)
- *ACM-BCB*: "Identifying Clock Gene Signatures Through Gene Expression Analysis for Cancer Therapy"
- *ICBCB (IEEE)*: "Identifying Neural Patterns and Biomarkers of ASD through Multi-phase Resting-State Functional MRI Analysis," DOI: 10.1109/ICBCB57893.2023.10246705
- 2-time Regeneron International Science and Engineering Fair (ISEF) Grand Award
- Regeneron Science Talent Search Top 300 Scholar
- British Biology Olympiad Gold Medalist, USA Biology Olympiad National Semifinalist
- 4-time American Invitational Mathematics Exam (AIME) Qualifier, Math Prize for Girls (MP4G)